

Group Problem Set 17: Work, Energy and Power

1. An object of mass 10 kg moves in a straight line and experiences a force of $F_x = (-10 \frac{N}{m})x^2$.

- What is the work done by this force as the object moves from $x = 1m$ to $x = 5m$?
- At $x = 1m$ the object has a speed of $10m/s$. What is the object's mass?
- What is the speed of the object when it reaches $x = 5m$?
- How far from its initial position, is the object when it stops?
- Using excel, can you plot the force v. distance?
- How much work is done within the first 1 meter?
- How much work is done on the object in the last 1 meter?

$$a) \int_1^5 -10x^2 dx = -\frac{10}{3} x^3 \Big|_{x=1}^{x=5} = -\frac{10}{3} (125) + \frac{10}{3} = -\frac{1240}{3} J$$

$$b) \text{ mass} = 10 \text{ kg}$$

$$c) K_1 = \frac{1}{2} m v_1^2 = 500 J$$

$$K_5 = 500 J + W_{1 \rightarrow 5} = 500 - \frac{1240}{3} = \frac{1500 - 1240}{3} = \frac{260}{3} J$$

$$v_5 = \sqrt{\frac{2K_5}{m}} = \sqrt{\frac{52}{3}} \text{ m/s}$$

$$d) 500 + \int_1^{x_{\text{end}}} -10x^2 dx = 0 \quad -\frac{10}{3} x^3 \Big|_{x=1}^{x=b} = -500$$

$$-\frac{10}{3} b^3 + \frac{10}{3} = -500$$

$$\frac{10}{3} b^3 = \frac{1510}{3}$$

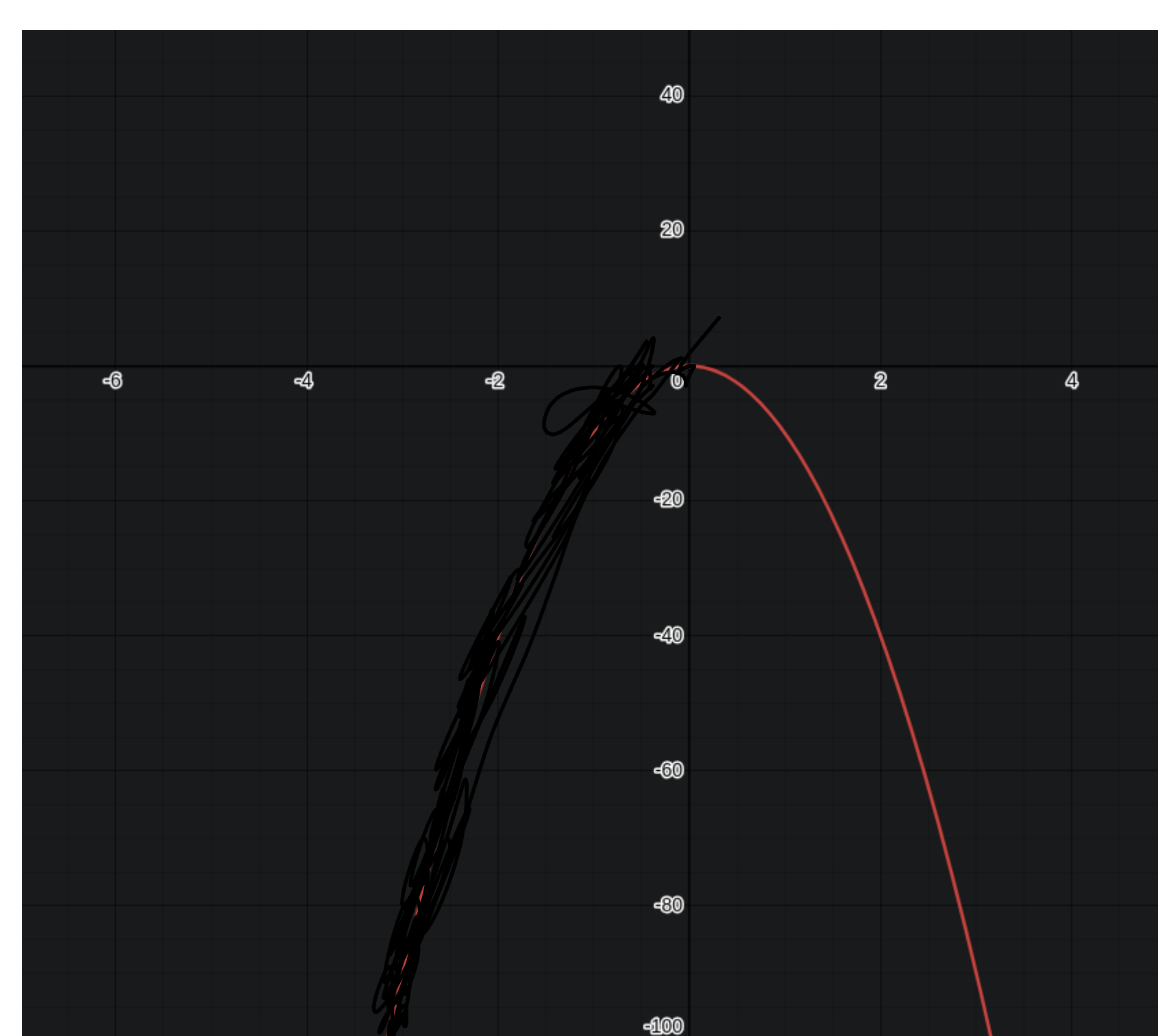
$$b^3 = 151$$

$$b = \sqrt[3]{151} = 5.33 \text{ m}$$

assuming $x=1m$ is initial, total displacement is 4.33m

e)

force



Distance

$$f) \int_0^1 -20x^2 dx = \left. -\frac{20}{3}x^3 \right|_0^1 = -\frac{20}{3} \text{ J}$$

$$g) \int_{\sqrt[3]{251}-1}^{\sqrt[3]{251}} -20x^2 dx = \left. -\frac{20}{3}x^3 \right|_0^1 = -\frac{20}{3}(251) + \frac{20}{3}(\sqrt[3]{251}-1)^3 = -235.49 \text{ J}$$